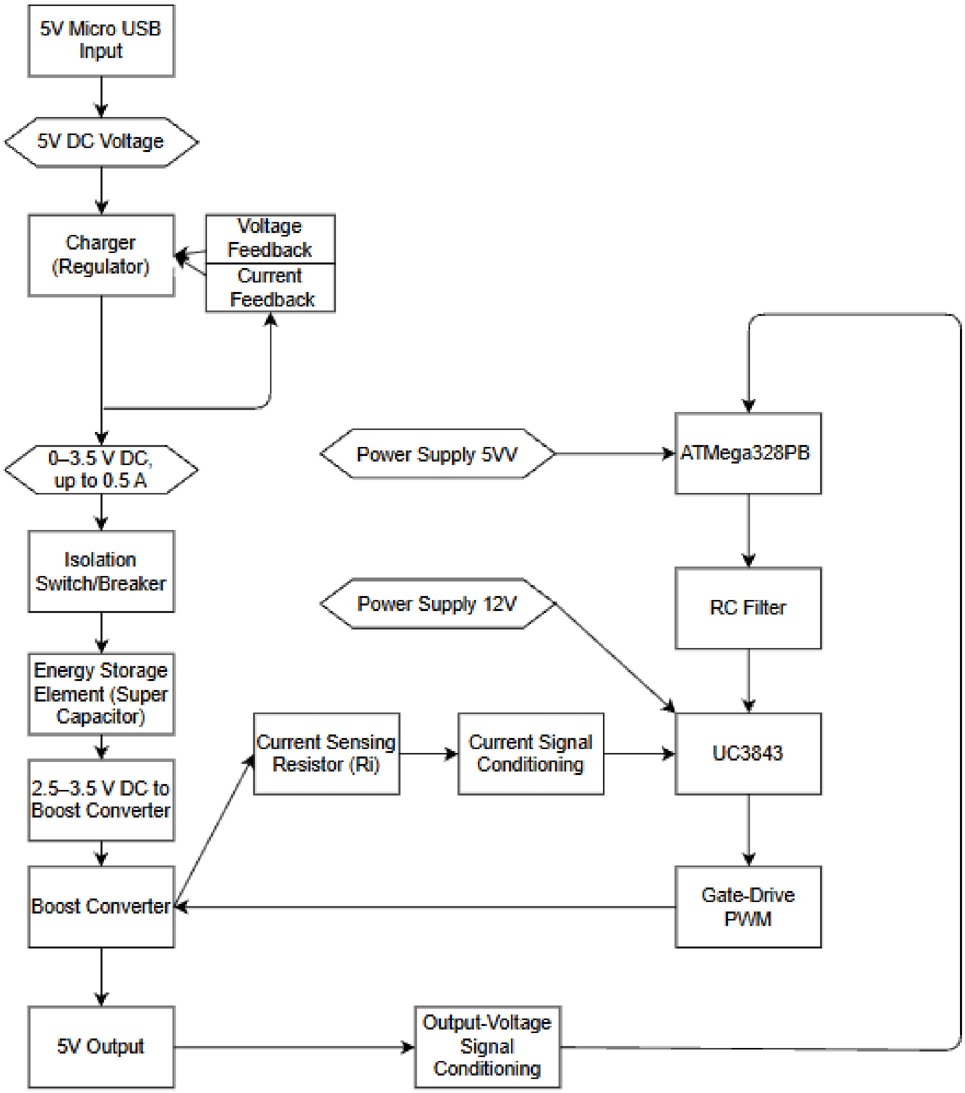


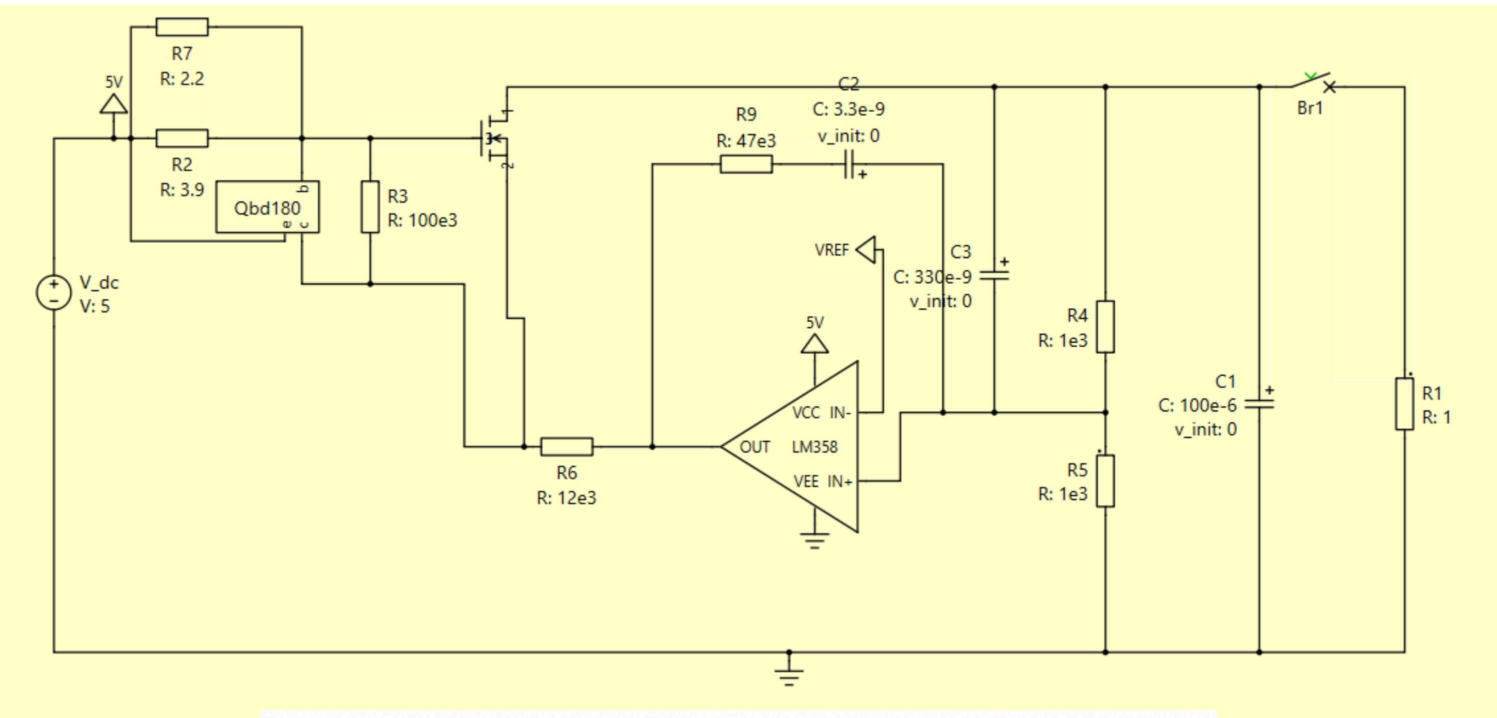
Parts search:

- QBD180 PNP BJT
- Irf9z24n P-Type Mosfet
- LM358 OPAMP
- Super Capacitor 30F or 150F
- UC3843
- ATMega328P

The IRF9Z24N P-channel MOSFET was selected as the switching device because it provides low conduction losses and simple gate driven requirements in a positive supply system. Compared with a BJT, the MOSFET requires negligible steady-state gate current, improving efficiency and reducing the loading on the control circuit. Its low on-resistance minimises voltage drop and power loss during charging and discharging, while its current and voltage ratings provide sufficient margin for the expected operating conditions.

A supercapacitor was selected as the primary energy storage element because it offers extremely high capacitance, rapid charge/discharge capability, and a significantly longer cycle life than rechargeable batteries. The supercapacitor can deliver large transient currents without substantial degradation, making it suitable for a portable power bank application requiring frequent charging and discharging.
The 150 F supercapacitor was selected to maximise stored energy and extend the operating time of the power bank. Although it requires a longer charging period and higher charging current, it better demonstrates the capability of the charging circuit and provides a more practical energy reserve.

P-type equivalent of BD179 BJT which is the BD180 was also used in the current limiting circuit within the charger.



Charger Specifications

Parameter	Value
Input Voltage Range	2.5 V_{in} to 3.5 V_{in}
Output Voltage	5 V_{DC}
Load Regulation	5%
Output Power	0.5 W to 3 W
Output Voltage Ripple	< 50 mV
Switching Frequency	100 kHz
R_{load}	50 ohm

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Current limiting:

if input current > 0.5A

BJT on when $V_E = V_B + 0.7V (V_{R_2})$

V_{R_2} = Input current × Resistance

$0.7 = I_{in} \times (2.2\Omega \| 3.9\Omega)$

$I_{in} = 0.7/1.4 = 0.5A$

R_3 : Pull up resistor

Voltage feedback divider:

$V_{fb} = 1.75V$

$V_{fb} = V_{out} \frac{R5}{R4+R5}$

$V_{fb} = V_{out} \frac{1k}{2k}$

$V_{fb} = \frac{V_{out}}{2}$

$V_{out} = 3.5V$

Opamp Loop:

LM358 max output = 3.5V

according to the datasheet

Compensation Components:

$R9 = 47k$, $C2 = 3.3 \times 10^{-9}$

$f = \frac{1}{2\pi RC} \rightarrow \frac{1}{2\pi \times 47k \times 3.3 \times 10^{-9}}$

$f = 1.03kHz$

Mosfet:

Mosfet is on when Source > Gate

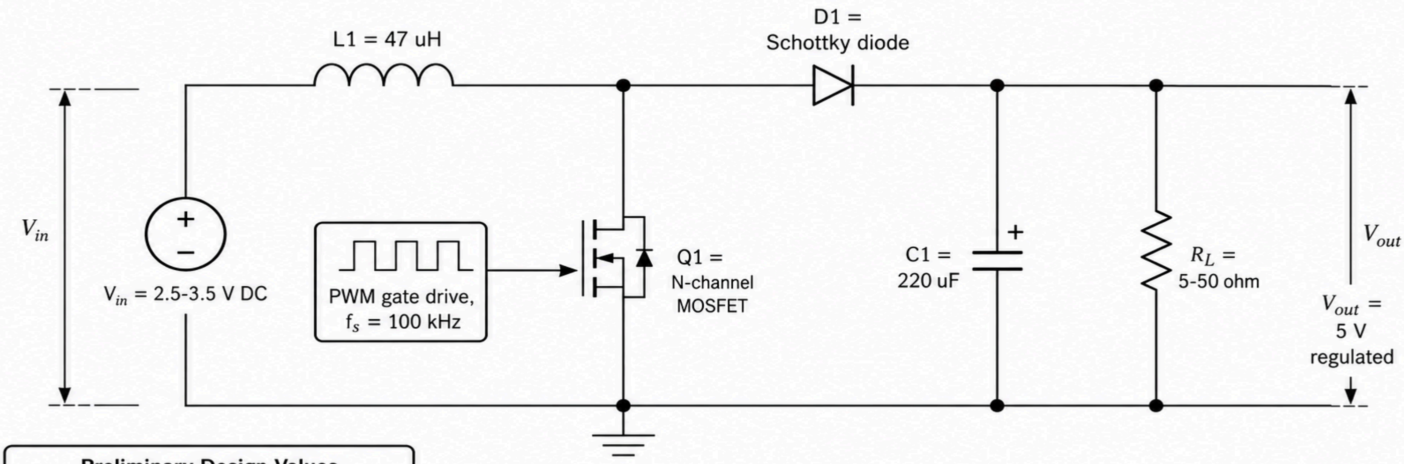
Source = 5V (4.3 if current limiting)

Gate = 3.5V from opamp max output

Drain = 3.5V from Vref voltage divider

Acts as a variable resistor with 1.5V drop

Preliminary Boost Converter Schematic



Preliminary Design Values

Output power	:	0.5-5 W
Output current	:	0.1-1.0 A
Duty cycle range	:	D = 0.30-0.50
Minimum load	:	5 ohm
Maximum load	:	50 ohm
Target ripple	:	< 50 mV

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a conventional non-isolated boost converter is proposed to convert the hybrid supercapacitor voltage of 2.5-3.5V to a regulated 5V output. The MOSFET stores energy in L1 during its ON interval and during its OFF interval, L1 transfers energy through D1 to C1 and the load R1

- $V_{out} = \frac{V_{in}}{1-D}$ $\quad D = 1 - \frac{V_{in}}{V_{out}}$

$D(2.5v) = 0.5$, $D(3.5v) = 0.3$

$R_{load} = \frac{V^2_{out}}{P_{out}} = 5 - 50\,\Omega$

$L_{min} = \max\left[\frac{R_{load} D(1-D)^2}{2 f_s}\right] \approx 37\,\mu H$

$L = 47\,\mu H$

- At $V_{in} = 2.5V$, $P_{out} = 5W$

$I_{out} = 1A$, $I_{in} = 2A$

$\Delta I_L = \frac{V_{in} D}{L f_s} = 0.266A$

$I_{L,peak} = I_{in} + \frac{\Delta I_L}{2} = 2.13A$

$C_{o,min} = \frac{I_{out} D}{f_s \Delta V_{out}} = 100\,\mu F$

- $C_o = 220\,\mu F$

$\Delta V_{out,ideal} = \frac{I_{out} D}{f_s C_o} = 22.7mV < 50mV$

Ideal PLECS simulation verifies the topology and initial component values. Closed loop UC3843/ ATmega control and non-ideal device losses will be incorporated during detailed design

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Task Division:
High level system schematic - Bruce/Kyle
Part search & Feasibility study - Alex\Bruce
Literature survey - Bruce/Alex
Calculations/diagrams - Saketh/Kyle